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Starter (Pre-requisites)

1) Write each of the following in the form $(x+p)^2+q$:
   a) $x^2+10x+28$ $(x+5)^2+3$
   b) x^2-6x+1 $(x-3)^2-8$
   c) x^2-12x $(x-6)^2-36$
   d) x^2+7x $\left(x+\frac{7}{2}\right)^2-\frac{49}{4}$

2) Find the equation of the line passing through each of the following pairs of points:
   a) $A(0,-6)$ and $B(4,3)$ $y=\frac{9}{4}x-6$
   b) $P(7,-5)$ and $Q(-9,3)$ $y=-\frac{1}{2}x-\frac{3}{2}$
   c) $R(-4,-2)$ and $T(5,10)$ $y=\frac{4}{3}x+\frac{10}{3}$

3) Use the discriminant to determine whether the following have two real solutions, one real solution or no real solutions.
   a) $x^2-7x+14=0$ $b^2-4ac=-7<0$ No real solutions
   b) $x^2+11x+8=0$ $b^2-4ac=193>0$ Two real solutions
   c) $4x^2+12x+9=0$ $b^2-4ac=0$ One real solution

4) Find the equation of the line that passes through the point $(3,-4)$ and is perpendicular to the line with equation $6x-5y-1=0$ $y=-\frac{5}{6}x-\frac{3}{2}$

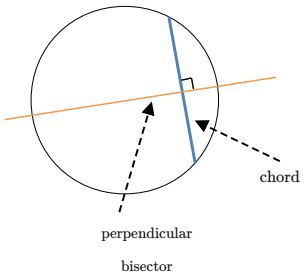
1: Equation of a circle

2: Intersections of lines + circles

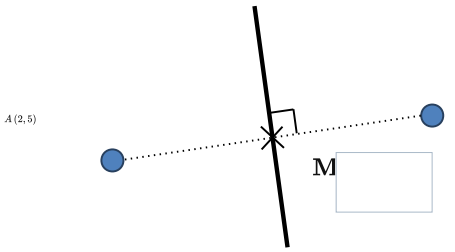
3: Chords, tangents and perpendicular bisectors.

4: Circumscribing Triangles

Later in the chapter you will need to find the perpendicular bisector of a chord of a circle.



What two properties does a perpendicular bisector of two points A and B have?



Just find the mean of the x values and the mean of the y values.

1. It passes through the midpoint of AB.
2. It is perpendicular to AB.

Equation?

m

m

Find the perpendicular bisector of the line AB where A and B have the coordinates:

a) $A\,(4, 7), B\,(10, 17)$

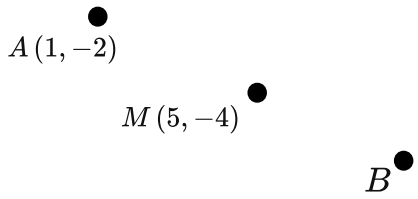
b) $A\,(x_1, y_1), B\,(x_2, y_2)$

a

b

Note: Do not try to memorise this!

A line segment AB is the diameter of a circle with centre $(5, -4)$. If A has coordinates $(1, -2)$, what are the coordinates of B?



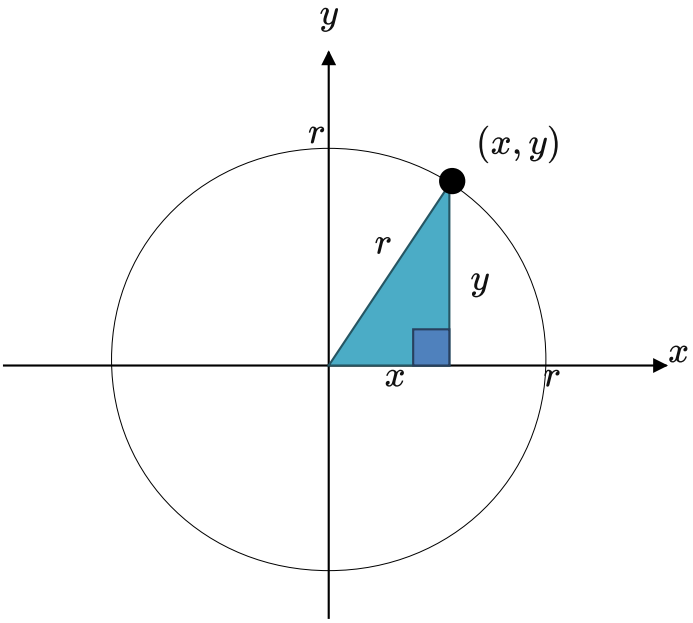
6 The line segment JK is a diameter of a circle, where J and K are $(\frac{3}{4}, \frac{4}{3})$ and $(-\frac{1}{2}, 2)$ respectively. The centre of the circle lies on the line segment with equation $y = 8x + b$. Find the value of b.

7 The line segment AB is a diameter of a circle, where A and B are $(0, -2)$ and $(6, -5)$ respectively. Show that the centre of the circle lies on the line $x - 2y - 10 = 0$.

2 The line FG is a diameter of the circle centre C, where F and G are $(-2, 5)$ and $(2, 9)$ respectively. The line l passes through C and is perpendicular to FG. Find the equation of l. (7 marks)

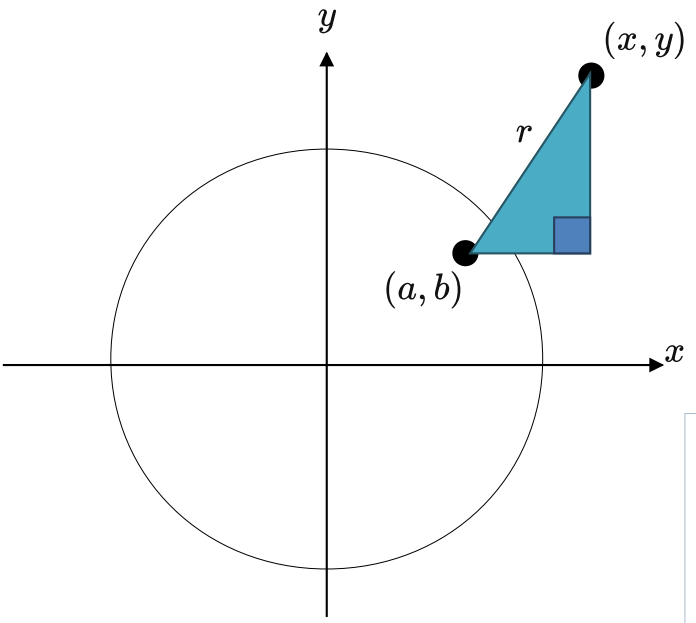
3 The line JK is a diameter of the circle centre P, where J and K are $(0, -3)$ and $(4, -5)$ respectively. The line l passes through P and is perpendicular to JK. Find the equation of l. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

5 Point X has coordinates $(7, -2)$ and point Y has coordinates $(4, q)$. The perpendicular bisector of XY has equation $y = 4x + b$. Find the value of q and the value of b.



Recall that a line can be a set of points (x, y) that satisfy some equation. Suppose we have a point (x, y) on a circle centred at the origin, with radius r. What equation must (x, y) satisfy?

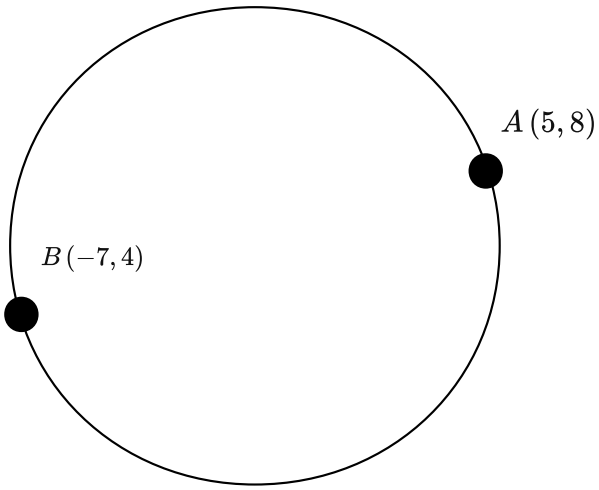
(Hint: draw a right-angled triangle inside your circle, with one vertex at the origin and another at the circumference)



Now suppose we shift the circle so it's now centred at (a, b). What's the equation now?

(Hint: What would the sides of this right-angled triangle be now?)

Centre	Radius	Equation
$(0,0)$	5	
$(1,2)$	6	
		$(x+3)^2+(y-5)^2=1$
		$(x+5)^2+(y-2)^2=49$
		$(x+6)^2+y^2=16$
		$(x-1)^2+(y+1)^2=3$
		$(x+2)^2+(y-3)^2=8$



A line segment AB is the diameter of a circle, where A and B have coordinates $(5,8)$ and $(-7,4)$ respectively. Determine the equation of the circle.

Hint: What two things do we need to use the circle formula?

The points A and B have coordinates $(5, -1)$ and $(13, 11)$ respectively.
(a) Find the coordinates of the mid-point of AB. (2)
Given that AB is a diameter of the circle C,
(b) find an equation for C. (4)

a

When the equation of a circle is in the form $(x-a)^2 + (y-b)^2 = r^2$, we can instantly read off the centre (a, b) and the radius r.
But what if the equation wasn't in this form?

Find the centre and radius of the circle with equation $x^2 + y^2 - 6x + 2y - 6 = 0$

Hint: Have we seen a method in a previous chapter that allows us to turn a x^2 term and a x term into a single expression involving x?

Textbook Note: There's a truly awful method, initially presented in the textbook, that allows you to find the centre/radius without completing the square. Don't even contemplate using it.

The circle C with centre T and radius r has equation
$x^2 + y^2 - 20x - 16y + 139 = 0$
(a) Find the coordinates of the centre of C. (3)
(b) Show that $r = 5$. (2)

Given the generic equation of a circle $x^2 + y^2 + ax + by + c = 0$ find a formula to determine the co-ordinate of the centre and the radius of the circle.

5 The line PQ is the diameter of the circle, where P and Q are $(5, 6)$ and $(-2, 2)$ respectively. Find the equation of the circle. (5 ...

6 The point $(1, -3)$ lies on the circle $(x - 3)^2 + (y + 4)^2 = r^2$. Find the value of r.

8 a Show that $x^2 + y^2 - 4x - 11 = 0$ can be written in the form $(x - a)^2 + y^2 = r^2$, where a and r are numbers to be found. (2 marks)
b Hence write down the centre and radius of the circle with equati...
$x^2 + y^2 - 4x - 11 = 0$. (2 marks)

9 a Show that $x^2 + y^2 - 10x + 4y - 20 = 0$ can be written in the form $(x - a)^2 + (y - b)^2 = r^2$, where a, b and r are numbers to be found. (2 m ...
b Hence write down the centre and radius of the circle with equation
$x^2 + y^2 - 10x + 4y - 20 = 0$. (2 m ...

12 The point $P(7, -14)$ lies on the circle with equation $x^2 + y^2 + 6x - 14y = 17$. The point Q also lies on the circle such that PQ is a diameter. Find the coordinates of point Q.

13 T $(x-k)^2 + y^2 = 41$ $(3, 4)$ possible values of k.

1 [MAT 2009 1B] The point on the circle $x^2 + y^2 + 6x + 8y = 75$ which is closest to the origin, is at what distance from the origin?

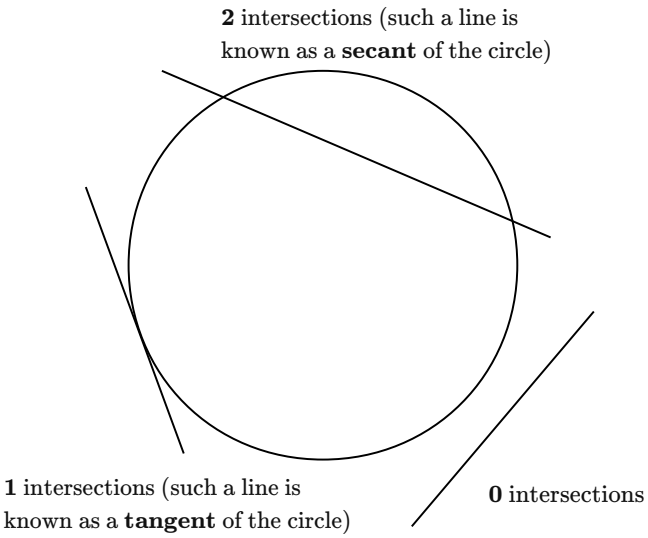
$(x+3)^2+(y+4)^2=100$

2 [MAT 2007 1D]
The point on the circle
$(x-5)^2 + (y-4)^2 = 4$
which is closest to the circle
$(x-1)^2 + (y-1)^2 = 1$
has what coordinates?

Drawing the circles on the same axes, and

3 [MAT 2016 1I] Let a and b be positive real numbers. If $x^2 + y^2 \leq 1$ then the largest that $ax + by$ can equal is what? Give your expression in terms of a and b.

Recall that to consider the **intersection of two lines**, we attempt to solve them **simultaneously** by substitution, potentially using the **discriminant** to show that there are no solutions (and hence no points of intersection).



Show that the line $y = x + 3$ never intersects the circle with equation $x^2 + y^2 = 1$.

Find the points of intersection where the line $y = x + 6$ meets $x^2 + (y - 3)^2 = 29$.

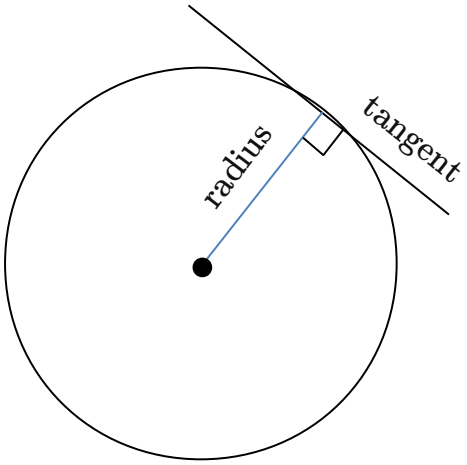
Using an algebraic (and not geometric) method, determine the k such that the line $y = x + k$ **touches** the circle with equation $x^2 + y^2 = 1$.

6 a Show that the line $x + y = 11$ meets the circle with equation $x^2 + (y - 3)^2 = 32$ at only one point.
b Find the coordinates of the point of intersection. (1 ma...

7 The line $y = 2x - 2$ meets the circle $(x - 2)^2 + (y - 2)^2 = 20$ at A and B.
a Find the coordinates of A and B.
b Show that AB is a diameter of the circle.

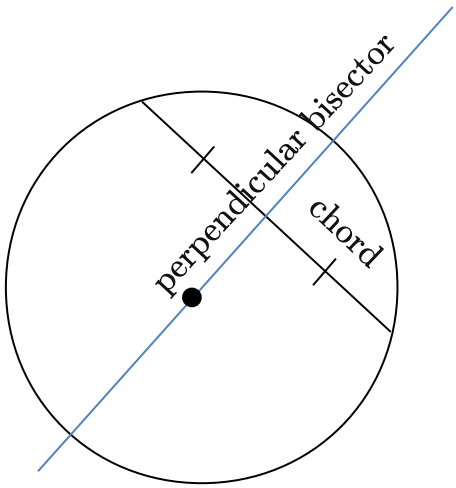
11 The line with equation $y = 4x - 1$ does not intersect the circle with equation $x^2 + 2x + y^2 = k$. Find the range of possible values of k.

There are two circle theorems that are of particular relevance to problems in this chapter, the latter you might be less familiar with:



The tangent is perpendicular to the radius (at the point of intersection).

Why this will help:
If we knew the centre of the circle and the point of intersection, we can easily find the gradient of the radius, and thus the gradient and hence equation of the tangent.



The perpendicular bisector of any chord passes through the centre of the circle.

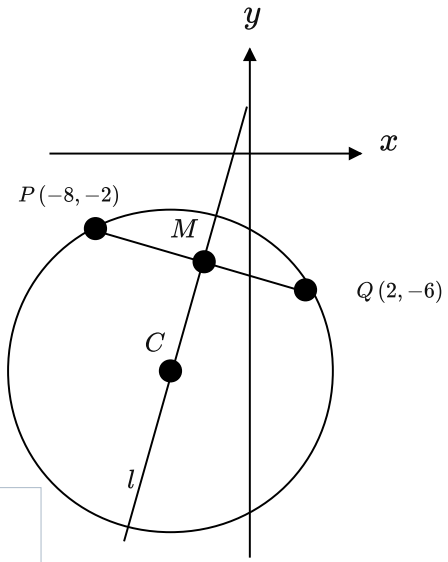
Why this will help:
The first thing we did in this chapter is find the equation of the perpendicular bisector. If we had two chords, and hence found two bisectors, we could find the point of intersection, which would be the centre of the circle.

The circle C has equation
$(x-3)^2+(y-7)^2=100$.
a) Verify the point $P\,(11,1)$ lies on C.
b) Find an equation of the tangent to C at the point P, giving your answer in the form $ax+by+c=0$

A circle C has equation
$(x-4)^2+(y+4)^2=10$
The line l is a tangent to the circle and has gradient -3. Find two possible equations for l, giving your answers in the form $y=mx+c$.

The points P and Q lie on a circle with centre C, as shown in the diagram. The point P has coordinates $(-8, -2)$ and the point Q has coordinates $(2, -6)$. M is the midpoint of the line segment PQ.
The line l passes through the points M and C.

a) Find an equation for l.
b) Given that the y-coordinate of C is -9:
   i) show that the x-coordinate of C is -5.
   ii) find an equation of the circle.



a

b

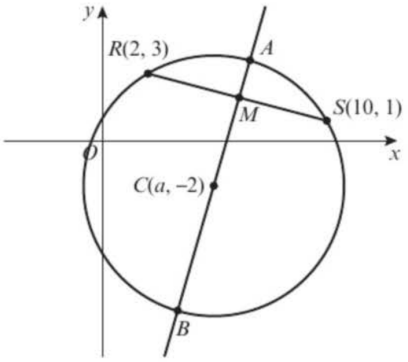
c

A circle has centre $C\,(3, 5)$, and goes through the point $P(6, 9)$. Find the equation of the tangent of the circle at the point P, giving your equation in the form $ax + by + c = 0$ where a, b, c are integers..

A circle passes through the points $A(0, 0)$ and $B\,(4, 2)$. The centre of the circle has x value -1. Determine the equation of the circle.

6 The tangent to the circle $(x+4)^2+(y-1)^2=242$ at $(7,-10)$ meets the y-axis at S and the x-axis at T.
a Find the coordinates of S and T. (5 marks)
b Hence, find the area of $\triangle OST$, where O is the origin. (3 marks)

10 The points R and S lie on a circle with centre $C(a,-2)$, as shown in the diagram.
The point R has coordinates $(2,3)$ and the point S has coordinates $(10,1)$.
M is the midpoint of the line segment RS.
The line l passes through M and C.
a Find an equation for l. (4 marks)
b Find the value of a. (2 marks)
c Find the equation of the circle. (3 marks)



d Find the points of intersection, A and B, of the line l and the circle. (5 marks)

Extension:

1 *[MAT 2012 1A]* Which of the following lines is a tangent to the circle with equation $x^2 + y^2 = 4$?

A) $x + y = 2$
B) $y = x - 2\sqrt{2}$
C) $x = \sqrt{2}$
D) $y = \sqrt{2} - x$

Mark scheme on next slide.

2 *[AEA 2006 Q4]* The line with equation $y = mx$ is a tangent to the circle C_1 with equation $(x + 4)^2 + (y - 7)^2 = 13$

(a) Show that m satisfies the equation $3m^2 + 56m + 36 = 0$

The tangents from the origin O to C_1 touch C_1 at the points A and B.

(b) Find the coordinates of the points A and B.

Another circle C_2 has equation $x^2 + y^2 = 13$. The tangents from the point $(4, -7)$ to C_2 touch it at the points P and Q.

(c) Find the coordinates of either the point P or the point Q.

Mark scheme on next slide.

3 *[STEP 2005 Q6]* (This is not a tangent/chord question but is worthwhile regardless!)

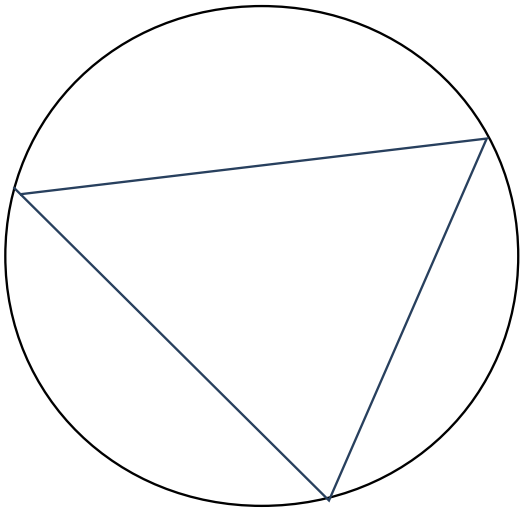
(i) The point A has coordinates $(5, 16)$ and the point B has coordinates $(4, -4)$. The variable P has coordinates (x, y) and moves on a path such that $AP = 2BP$. Show that the Cartesian equation of the path of P is $(x + 7)^2 + y^2 = 100$.

(ii) The point C has coordinates $(a, 0)$ and the point D has coordinates $(b, 0)$. The variable point Q moves on a path such that $QC = k \times QD$, where $k > 1$.

Given that the path of Q is the same as the path of P, show that $\frac{a+7}{b+7} = \frac{a^2+51}{b^2+51}$

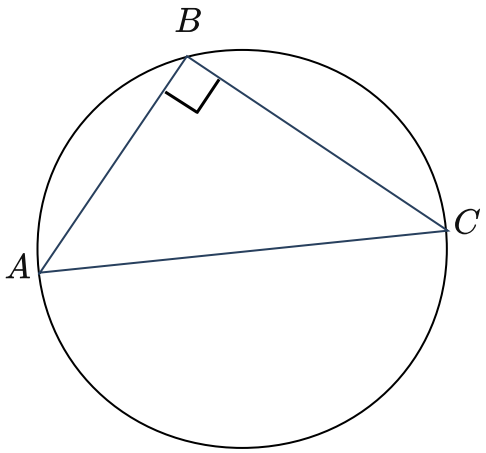
Show further that $(a + 7)(b + 7) = 100$, in the case $a \neq b$.

(c)

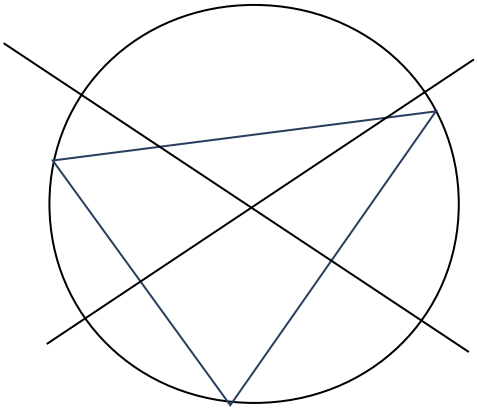


We'd say:

- The triangle **inscribes** the circle.
  (A shape inscribes another if it is inside and its boundaries touch but do not intersect the outer shape)
- The circle **circumscribes** the triangle.
- If the circumscribing shape is a circle, it is known as the **circumcircle** of the triangle.
- The centre of a circumcircle is known as the **circumcentre**.



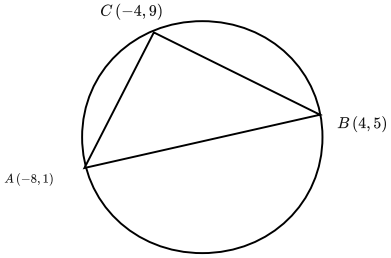
If $\angle ABC = 90°$ then:



Given three points/a triangle we can find the centre of the circumcircle by:

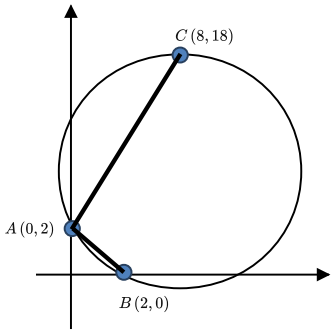
- **$\angle$ABC=90° therefore AB is perpendicular to BC.**

[Textbook] The points $A\,(-8,1), B\,(4,5), C\,(-4,9)$ lie on a circle.
a) Show that AB is a diameter of the circle.

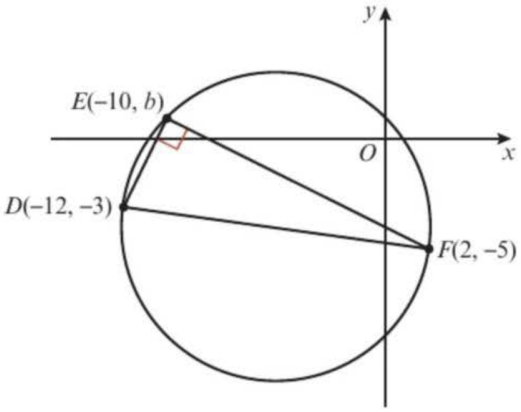


b) Hence find the equation of the circle.

The points $A\,(0,2), B\,(2,0), C\,(8,18)$ lie on the circumference of a circle. Determine the equation of the circle.



8 The points $D(-12,-3), E(-10,b)$ and $F(2,-5)$ lie on the circle C as shown in the diagram.
Given that $\angle DEF = 90^\circ$ and $b > 0$
a show that $b = 1$ (5 marks)
b find an equation for C. (4 marks)



9 A circle has equation $x^2 + 2x + y^2 - 24y - 24 = 0$.
a Find the centre and radius of the circle.
b The points $A(-13, 17)$ and $B(11, 7)$ both lie on the circumference of the circle. Show that AB is a diameter of the circle.
c The point C lies on the negative x-axis and the angle $ACB = 90°$. Find the coordinates of C.

Extension:

1 *[STEP 2009 Q8 Edited]* If equation of the circle C is $(x - 2t)^2 + (y - t)^2 = t^2$, where t is a positive number, it can be shown that C touches the line $y = 0$ as well as the line $3y = 4x$.

Find the equation of the incircle of the triangle formed by the lines $y = 0$, $3y = 4x$ and $4y = 3x + 5$.

Note: The incircle of a triangle is the circle, lying totally inside the triangle, that touches all three sides.

Solution: $\mathbf{(x-2)^2+(y-1)^2=1}$